

Randy Tang

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SUMMARY

Software engineer with 4+ years building and optimizing high-throughput, production-critical data pipelines processing tens of millions of daily transactions. Deep Python proficiency, hands-on API design and orchestration experience, and strong foundations in concurrency, retry logic, and error handling. Cornell CS graduate with growing experience in LLM APIs and generative AI tooling.

TECHNICAL SKILLS

Languages: Python, SQL, Java (fundamentals) | **Frameworks/Infra:** FastAPI (familiar), Flask, Docker, Postgres, SQLite, Redis (exposure), Git, GitHub Actions (exposure)

Data/Systems: High-throughput pipelines, concurrency, AMPS message queues, Pandas, API design & orchestration, retries & error handling

AI / ML: LLM APIs, agentic coding & orchestration, ML data processing & annotation | **Practices:** Performance optimization, pipeline design, code review, documentation

EXPERIENCE

Software Engineer — Bank of America

Aug 2022 – Present

Cirrus OTC Trade Reporting | Python, Pandas, Quartz, AMPS

- Designed and built **production orchestration** across multiple services processing **tens of millions of transactions daily**, with retries, error handling, and load balancing.
- Built an extensible **reference-data framework**: pipeline of services connected via message queues, querying multiple external APIs per event and handling failures gracefully.
- Implemented a **T+1 Unique Trade Identifier service** validating trade-ID uniqueness across counterparties and issuing corrections at millions of transactions per day.
- Developed a parallelized **high-volume backreporting service**, automating a previously manual process with substantial speed and throughput gains.
- Designed clean **API interfaces** between services: internal model transformer (to outgoing XML) and a response parser to update transaction statuses.
- Optimized reliability, latency, and throughput of core services by orders of magnitude; authored Python/Pandas tooling to analyze 50–100GB datasets.

Data Analyst (Contract) — Microsoft

Mar 2022 – Aug 2022

Viva Topics | Python, Pandas, Jupyter

- Processed and annotated data used to **train ML models**; built Python pipelines to clean, transform, and report on user-collected datasets for the data science team.

Engineering Intern — Northrop Grumman

Fall 2019, Summer 2020

NX, ANSYS

- Designed and released 50+ engineering drawings including flight and test components; performed structural analyses for large assemblies and fastener stacks.

PROJECTS

- Reading Recommendations App** (Flask, Gemini API, SQLite) — Built an **LLM-powered** daily recommender integrating generative AI APIs over a Project Gutenberg corpus, handling API orchestration and error handling.
- Market Data Dashboard** (Postgres, Docker, Twilio) — Ingests market data from multiple external APIs (Binance, Hyperliquid); automated alerting pipeline with SMS notifications on signal triggers.
- Polymarket Scanner** (Flask, Discord API, SQLite) — Scans Polymarket for large trades and unusual activity; lightweight API backend with real-time alert delivery.

EDUCATION

Cornell University, College of Engineering — Ithaca, NY

Sep 2017 – Dec 2021

B.S. Computer Science & B.S. Mechanical Engineering | GPA 3.7

- CUAIR (Cornell Unmanned Aircraft)** — Developed systems for an autonomous plane including obstacle avoidance, object classification, and airdrop; top-performing team at the AUVSI SUAS competition.